

PRODUCT LITERATURE

TUSSIFIED SYRUP

Each 5 ml contains

Triprolidine Hydrochloride USP	1.25 mg
Pseudoephedrine Hydrochloride USP	30 mg

Preservatives

Methyl Paraben BP	0.1% w/v
Propyl Paraben BP	0.01% w/v

Product Description

Orange coloured clear syrup with sweet taste and fruity odour.

Pharmacodynamic properties

Triprolidine provides symptomatic relief in conditions believed to depend wholly or partly upon the triggered release of histamine. It is a potent competitive histamine H¹-receptor antagonist of the pyrrolidine class with mild central nervous system depressant properties, which may cause drowsiness. Pseudoephedrine has direct and indirect sympathomimetic activity and is an effective upper respiratory tract decongestant. Pseudoephedrine is substantially less potent than ephedrine in producing both tachycardia and elevation of systolic blood pressure and considerably less potent in causing stimulation of the central nervous system.

After oral administration of a single dose of 2.5mg triprolidine to adults the onset of action, as determined by the ability to antagonise histamine-induced weals and flares in the skin, is within 1 to 2 hours. Peak effects occur at about 3 hours and, although activity declines thereafter, significant inhibition of histamine-induced weals and flares still occurs 8 hours after the dose. Pseudoephedrine produces its decongestant effect within 30 minutes, persisting for at least 4 hours.

Pharmacokinetics properties

After the administration of 2.5 mg triprolidine hydrochloride and 60 mg pseudoephedrine hydrochloride in healthy adult volunteers, the peak plasma concentration (C_{max}) of triprolidine is approximately 5.5 ng/ml - 6.0 ng/ml, occurring at about 2.0 hours (T_{max}) after drug administration. The plasma half life of triprolidine is approximately 3.2 hours. The C_{max} of pseudoephedrine is approximately 180ng/ml with T_{max} approximately 2.0 hours after drug administration. The plasma half life of pseudoephedrine is approximately 5.5 hours (urine pH maintained between 5.0-7.0). The plasma half life of pseudoephedrine is markedly decreased by acidification of urine and increased by alkalinisation.

Indication

For the temporary relief of nasal and sinus congestion associated with the common cold and allergic rhinitis.

Recommended Dose

To be taken 3 or 4 times daily:

Adult: 10 ml,
6 - 12 years: 5 ml,
2 - 6 years: 2.5 ml.

Route of Administration

Oral

Contraindications

Tussified Syrup is contra-indicated in individuals who have previously exhibited intolerance to it or to pseudoephedrine or triprolidine.

Tussified Syrup is contra-indicated in patients who are taking or have taken monoamine oxidase inhibitors within the preceding two weeks. The concomitant use of pseudoephedrine and this type of product may occasionally cause a rise in blood pressure.

Tussified Syrup is contra-indicated in patients with severe hypertension or severe coronary artery disease.

The antibacterial agent furazolidone is known to cause a dose-related inhibition of monoamine oxidase. Although there are no reports of hypertensive crises caused by the concurrent administration of Tussified Syrup and furazolidone they should not be taken together.

Warnings and Precautions

- When used for treatment of cough and cold:
 - (i) Not to be used in children less than 2 years of age.
 - (ii) To be used with caution and doctor's/ pharmacist's advice in children 2 to 6 years of age.
- Tussified Syrup may cause drowsiness and impair performance in tests of auditory vigilance. Patients should not drive or operate machinery until they have determined their own response. If affected avoid driving vehicles or other machinery.
- Although there are no objective data, users of Tussified Syrup should avoid the concomitant use of alcohol or other centrally acting sedatives.
- Although pseudoephedrine has virtually no pressor effect in normotensive patients, Tussified Syrup should be used with caution in patients taking anti-hypertensive agents, tricyclic antidepressants or other sympathomimetic agents, such as decongestants, appetite suppressants and amphetamine-like psychostimulants. The effects of a single dose of Tussified Syrup on the blood pressure of these patients should be observed before recommending repeated or unsupervised treatment.
- As with other sympathomimetic agents Tussified Syrup should be used with caution in patients with hypertension, heart disease, diabetes, hyperthyroidism, elevated intraocular pressure and prostatic enlargement.
- There have been no specific studies of Tussified Syrup in patients with hepatic and/or renal dysfunction. Caution should be exercised in the presence of severe renal or hepatic impairment.
- There is insufficient information available to determine whether Triprolidine or pseudoephedrine have mutagenic or carcinogenic potential.
- Systemic administration of pseudoephedrine in rats, up to 7 times the human daily dosage in females and 35 times the human daily dosage in males, did not impair fertility nor alter foetal morphological development and survival.
- No studies have been conducted in animals to determine if Triprolidine has the potential to impair fertility.
- There is no information on the effects of Tussified Syrup on human fertility.

Interactions with Other Medicament

Concomitant use of Tussified Syrup with sympathomimetic agents, such as decongestants, tricyclic antidepressants, appetite suppressants and amphetamine-like psychostimulants, or with monoamine oxidase inhibitors which interfere with the catabolism of sympathomimetic amines, may occasionally cause a rise in blood pressure.

Because of its pseudoephedrine content, Tussified Syrup may partially reverse the hypotensive action of drugs which interfere with sympathetic activity including bretylium, betanidine, guanethidine, debrisoquine, methyl dopa, alpha-and beta-adrenergic blocking agents.

Pregnancy and Lactation

Although pseudoephedrine and triprolidine have been in widespread use for many years without apparent ill consequence, there are no specific data on their use during pregnancy. Caution should therefore be exercised by balancing the potential benefit of treatment to the mother against any possible hazards to the developing foetus.

Systemic administration of triprolidine in rats and rabbits up to 75 times the human dose did not produce teratogenic effects.

Systemic administration of pseudoephedrine, up to 50 times the human daily dosage in rats and up to 35 times the human daily dosage in rabbits, did not produce teratogenic effects.

Pseudoephedrine and triprolidine are excreted in breast-milk in small amounts but the effect of this on breast-fed infants is not known. It has been estimated that approximately 0.5 to 0.7% of a single dose of pseudoephedrine ingested by a mother will be excreted in the breast-milk over 24 hours.

Side Effects

Central nervous system depression or excitation may occur, drowsiness being reported most frequently. Sleep disturbance and, rarely, hallucinations have been reported.

Skin rashes, with or without irritation, tachycardia, and dryness of mouth, nose and throat, have occasionally been reported.

Urinary retention has been reported occasionally in men receiving pseudoephedrine; prostatic enlargement could have been an important predisposing factor.

Symptoms and Treatment of Overdose

The effects of acute toxicity from Tussifed Syrup may include drowsiness, lethargy, dizziness, ataxia, weakness, hypotonicity, respiratory depression, dryness of the skin and mucous membranes, tachycardia, hypertension, hyperpyrexia, hyperactivity, irritability, convulsions, and difficulty with micturition.

Necessary measures should be taken to maintain and support respiration and control convulsions. Gastric lavage should be performed up to 3 hours after ingestion if indicated. Catheterisation of the bladder may be necessary. If desired, the elimination of pseudoephedrine can be accelerated by acid diuresis or by dialysis.

Packing

Packed in plastic container of 100 ml

Pharmaceutical Precautions

Store below 30°C in a dry place, protected from direct light.

Keep away from reach of children.

Jauhkan daripada kanak-kanak

SHAKE WELL BEFORE USE

Expiry Date

2 years from date of manufacture

Name and Address of Product Registration Holder/ Distributor:

ZONTRON PHARMACEUTICALS SDN.BHD. (445695-T)

Lot 10 & 11, PERDA Industrial Park,

Lorong IKS Simpang Ampat B,

14100 Simpang Ampat, S.P.S.,

Pulau Pinang, Malaysia

Manufacturer by:

TERAPUTICS SDN. BHD. (590500-W)

Lot 10 & 11, PERDA Industrial Park,

Lorong IKS Simpang Ampat B,

14100 Simpang Ampat, S.P.S.,

Pulau Pinang, Malaysia

Malaysian Drug Registration No.

MAL19890435ACZ

Date of Revision: 08/11/2019